

EEE283

Digital Logic Design

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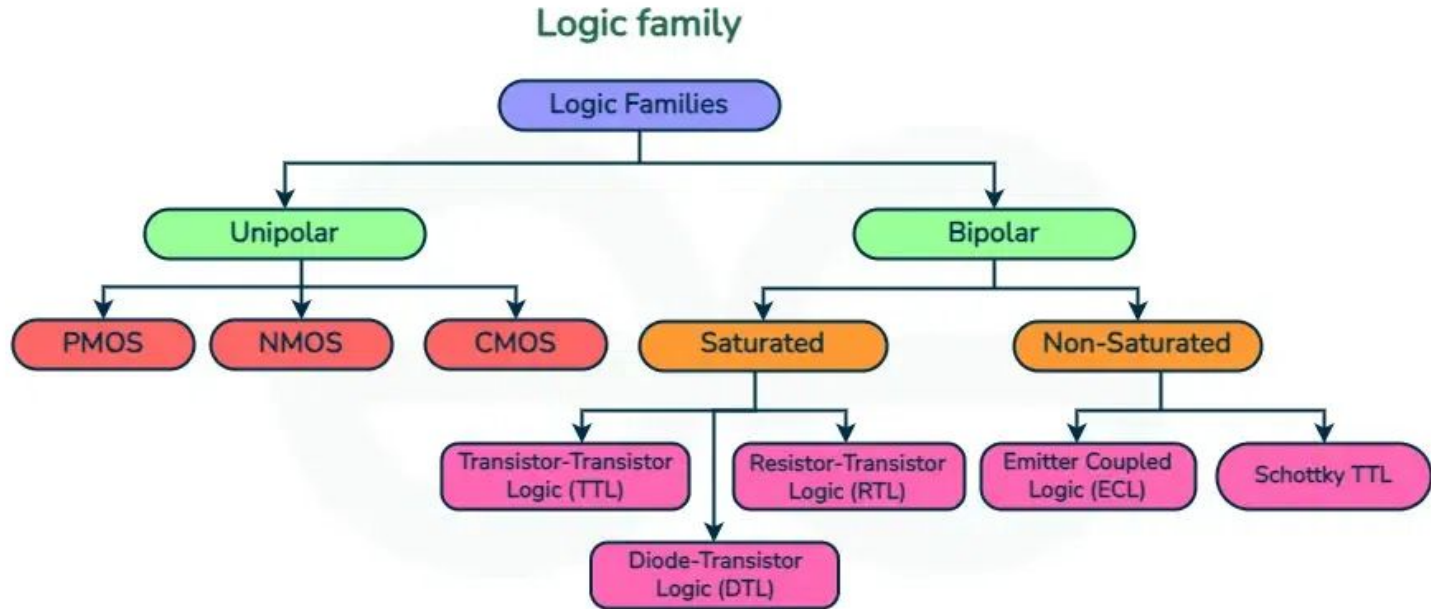
Topic 7: **Logic Family**

(adapted from Dr. A.S.M Mohsin's slides)

Logic Family

- **Definition:** Logic families are technologies used to build Logic. Same Logic Level, Supply Voltage, same Fabrication technology and Compatible ICs.
- **Logic Gates:** These are digital circuits (AND, OR, NOT, etc.) that perform basic logical operations within those families.
- **Logic Levels:** Systems use **Positive Logic** (High = 1, Low = 0) or **Negative Logic** (High = 0, Low = 1) to define binary states.
- **Components:** Gates are built using **diodes and transistors**, which act as electronic switches.
- **Selection Factors:** Choosing a family depends on performance needs like **speed, power use, and fan-out capacity**.

Classification



Unipolar Logic Family

Unipolar means having a single type of charge carrier. This logic family uses transistors that have either electrons or holes(not both) as charge carriers. The most commonly used technology is Metal-Oxide-Semiconductor(MOS). A common example is Complementary MOS (**CMOS**) logic.

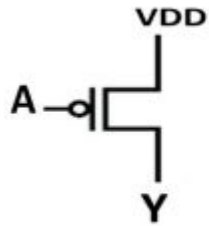


Fig-PMOS

A	Y
0	1
1	0

A=input
Y=output

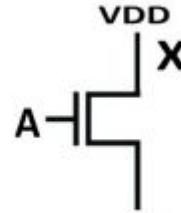


Fig-NMOS

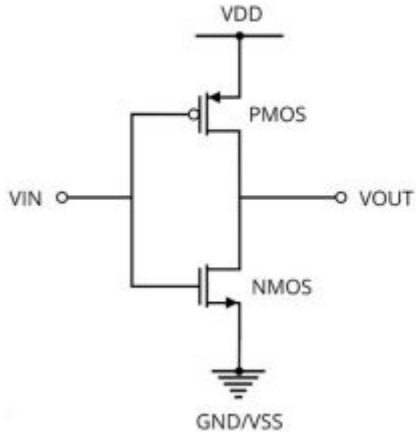
A	X
0	1
1	0

A=input
X=output

Voltage	PMOS	NMOS
0V (0)	ON	OFF
5V (1)	OFF	ON

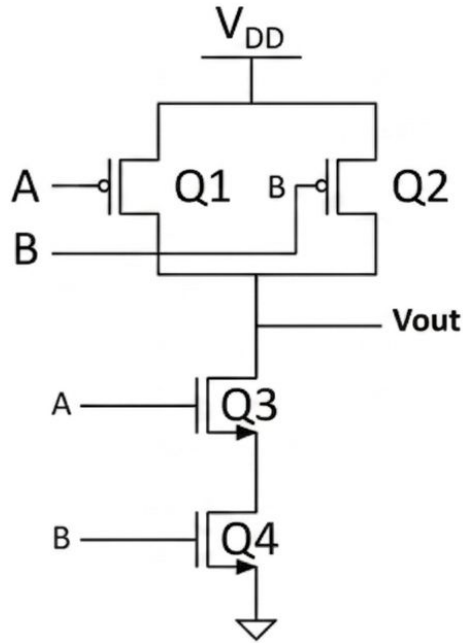
CMOS Inverter

C-MOS is a combination of P-MOS and N-MOS



Vin	PMOS(Q1)	NMOS(Q2)	Vout
0	ON	OFF	1
1	OFF	ON	0

CMOS NAND

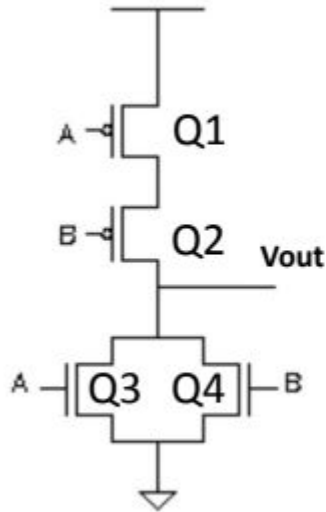


- PMOS –PULL UP
- NMOS-PULL DOWN

A	B	Q1	Q2	Q3	Q4	V_{out}
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

- PMOS: ON=0 & OFF= 1
- NMOS: ON=1 & OFF=0

CMOS NOR



- PMOS –PULL UP

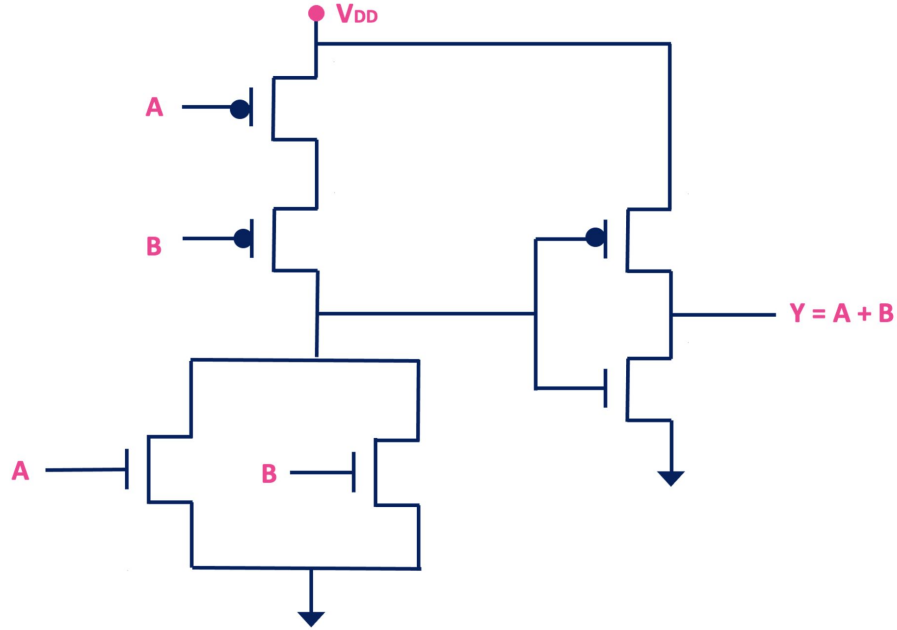
- NMOS-PULL DOWN

A	B	Q1	Q2	Q3	Q4	V_{out}
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	0
1	0	OFF	ON	ON	OFF	0
1	1	OFF	OFF	ON	ON	0

- PMOS: ON=0 & OFF= 1

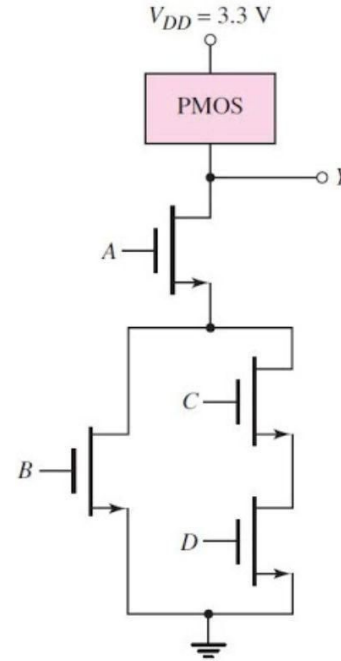
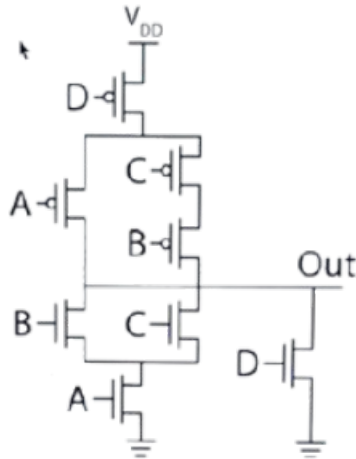
- NMOS: ON=1 & OFF=0

CMOS OR



Try CMOS AND on your own!

Logic Circuits using CMOS



Pros and Cons of CMOS

Advantages of CMOS

- Extremely low power consumption
- High fan-out (~50)
- Can operate in wider temperature ranges(-55 to 125 degree C)
- No static power dissipation. Power is dissipated only for switching MOSFETs.
- Best Noise Immunity

Disadvantages of CMOS

- Slow speed of operation
- Propagation delay time is around 50ns while this is around 10-12ns in TTL

Resistor Transistor Logic (RTL)

RTL-Basic Universal Gate: NOR

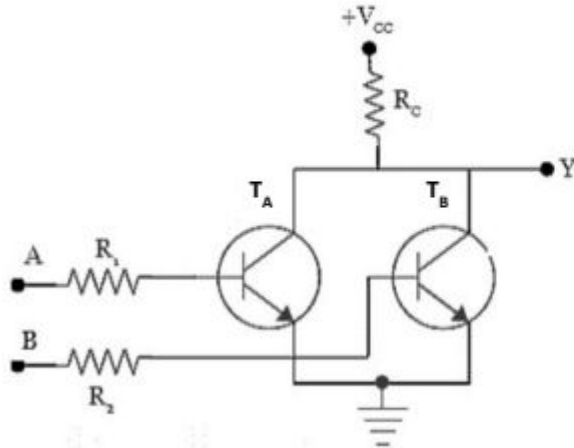


Fig-Two input NOR gate

When base (B) is high transistor (T) goes to saturation- **short circuit**

When base (B) is low transistor (T) goes to cutoff- **open circuit**

	A	B	T _A	T _B	V _{out} (Y)
Case-1	0	0	OPEN	OPEN	1
Case-2	0	1	OPEN	SHORT	0
Case-3	1	0	SHORT	OPEN	0
Case-4	1	1	SHORT	SHORT	0

RTL NAND

RTL-Basic Universal Gate: NAND

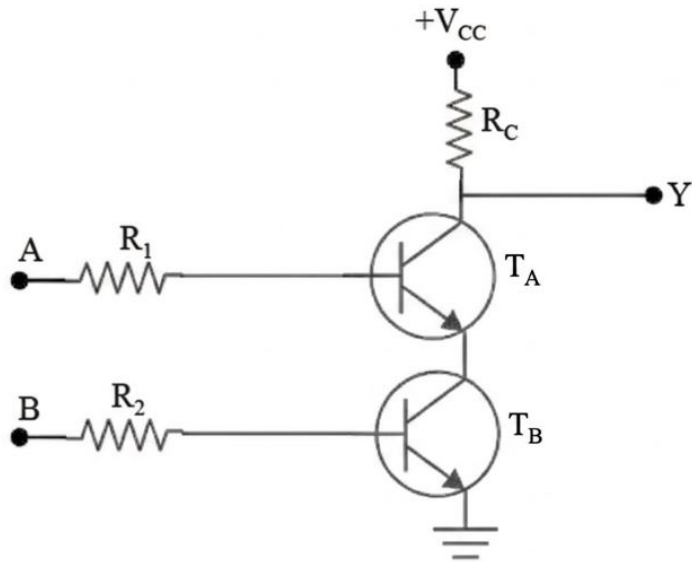


Fig-Two input NAND gate

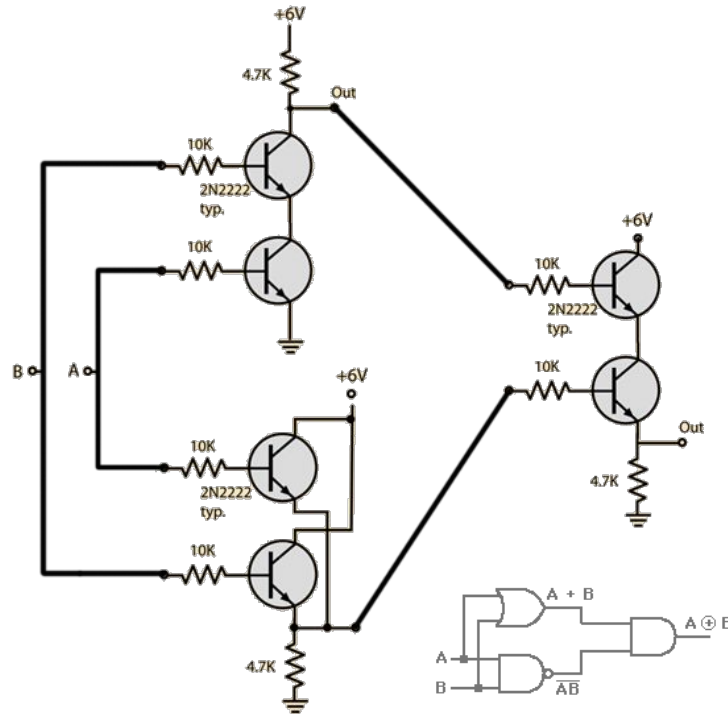
RTL NAND with series transistors T_A and T_B .

When base (A) and base (B) are both high, both transistors go to saturation (short) - **low** output.

If either base (A) or base (B) is **low**, at least one transistor is in cutoff (open) - high output.

	A	B	T_A	T_B	$V_{out}(Y)$
Case-1	0	0	OPEN	OPEN	1
Case-2	0	1	OPEN	SHORT	1
Case-3	1	0	SHORT	OPEN	1
Case-4	1	1	SHORT	SHORT	0

Logic Circuits using RTL



Pros and Cons of RTL

Advantages:

- The RTL circuit uses a least amount of transistors for combining various input signals, which helps in amplifying & inverting the combined resulting signal
- RTL gates are simple & inexpensive.
- These are handy due to both normal as well as inverted signals are frequently available.
- RTL is simple to design & less component count which makes it popular in digital electronics.
- Resistor transistor logic is replaced with very advanced logic families like TTL & CMOS because of their enhanced performance & efficiency.
- It reduces the usage of several semiconductor components.

Pros and Cons of RTL

Disadvantages:

- The resistor transistor logic has high current dissipation whenever the transistor behaves to overdrive the o/p biasing resistor.
- It has high power dissipation whenever the transistor is turned on by supplying current within the base & collector resistors.
- It has limited fan-in.
- These circuits' speed is quite slow as compared to other types of logic families because of the utilization of transistors & resistors.
- RTL circuits are complex.
- These circuits have poor noise immunity that makes them vulnerable to interference & degradation of signal.
- RTL circuits need fairly high voltage levels mainly for proper operation, which limits their compatibility with other systems.

Direct Coupled Transistor Logic (DCTL)

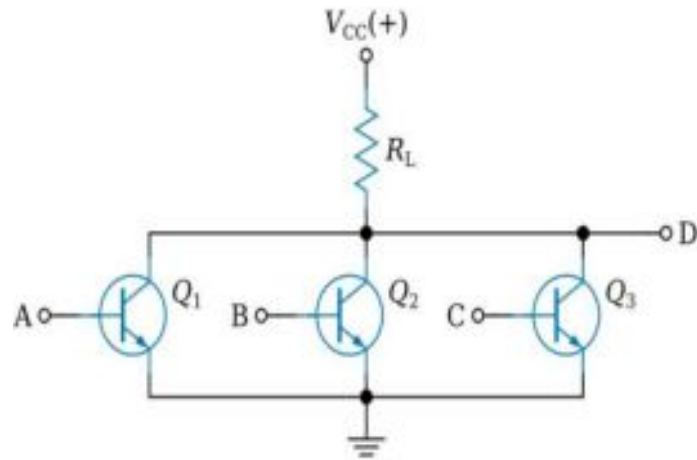


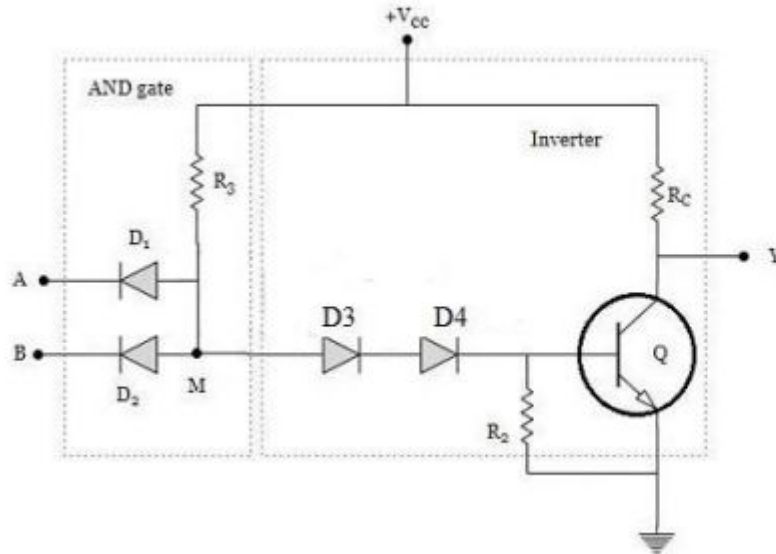
Fig- 3 input NOR

Difference Between RTL & DCTL -

Name	RTL (Resistor-Transistor Logic)	DCTL (Direct-Coupled Transistor Logic)
Input connection	Input applied to transistor base through resistors	Input applied directly to transistor base
Resistors	Required for base current limiting	Not required
Power consumption	Higher	Lower
Noise margin	Better (more reliable)	Poor (sensitive to variations)

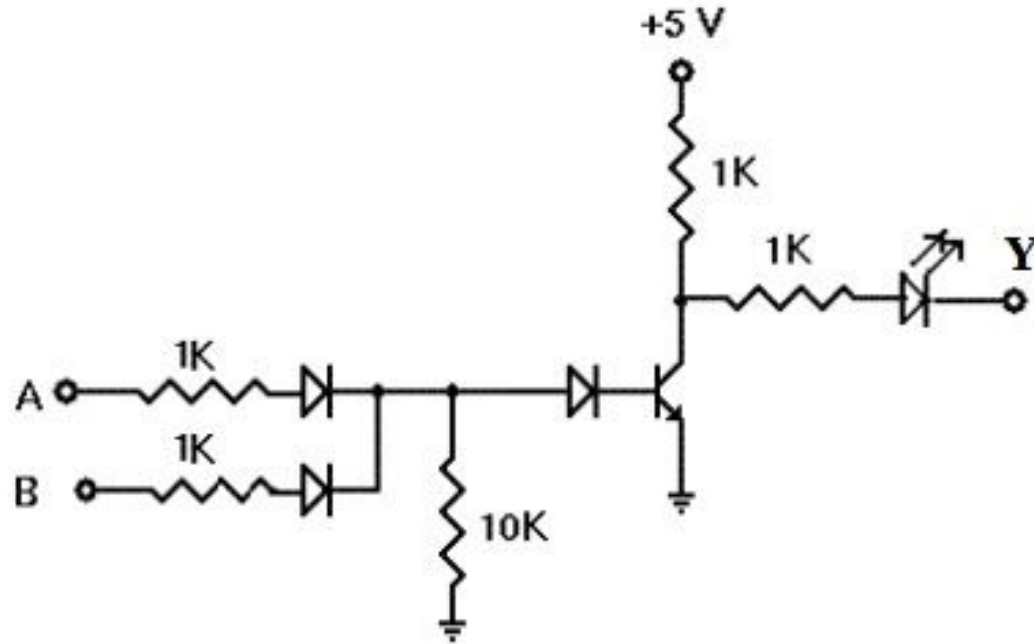
Diode Transistor Logic (DTL)

In Diode Transistor Logic, **diodes are used for AND and OR operations** while transistors are used for **logical inversion** and amplification. DTL is used to design and fabricate digital circuits that use diodes in the input stage and BJTs at the output stage

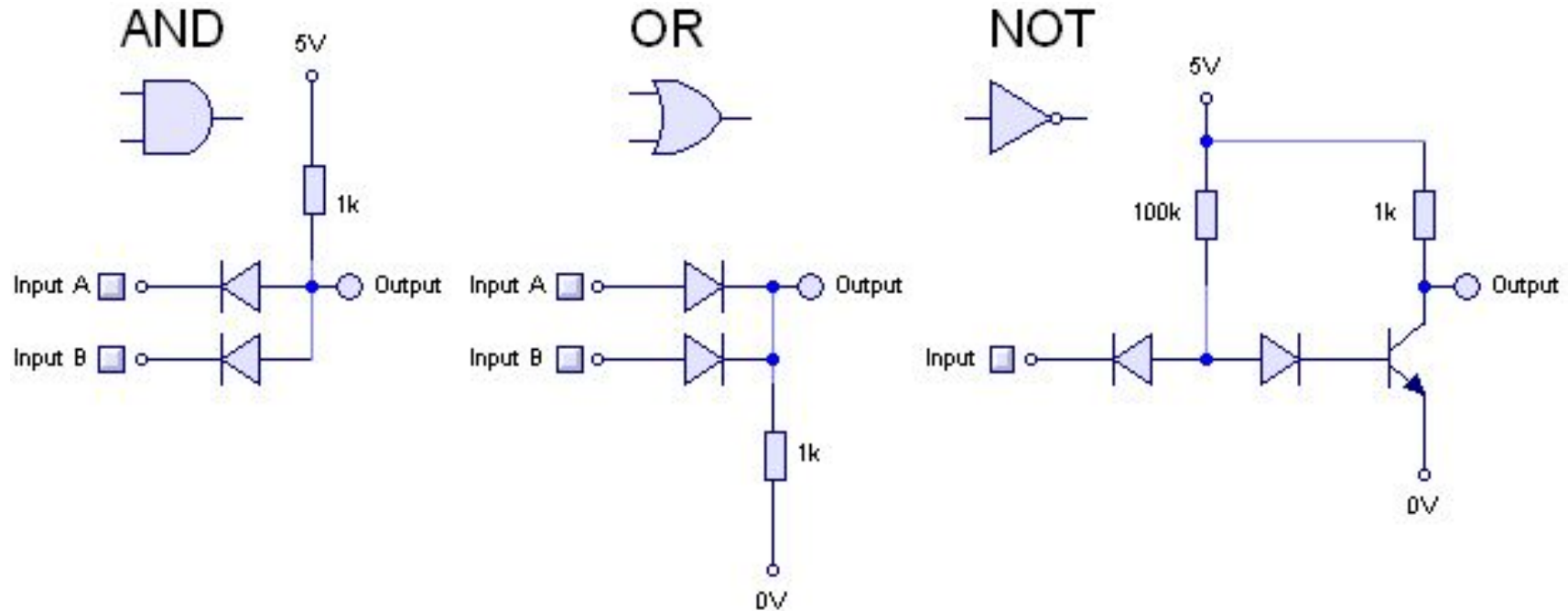


A	B	V _{out}
0	0	1
0	1	1
1	0	1
1	1	0

DTL NOR

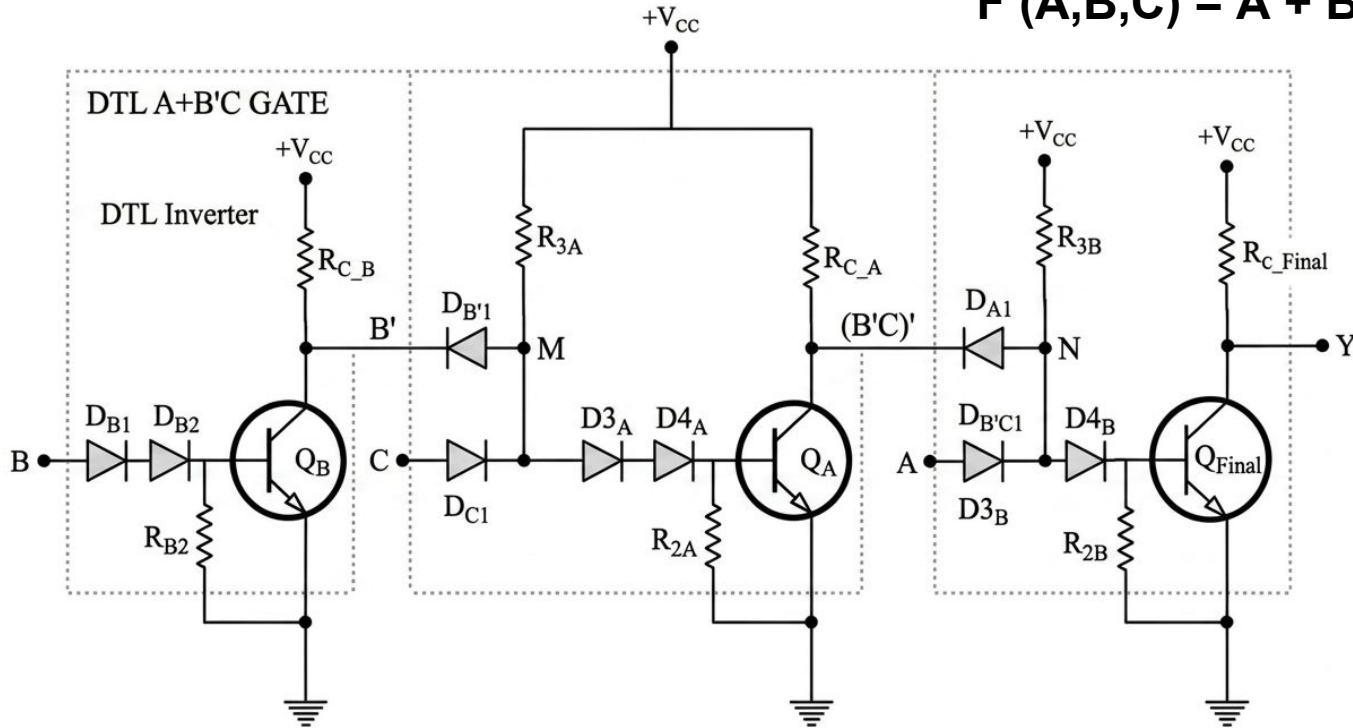


Basic Gates using DTL



Logic Circuits using DTL

$$F(A,B,C) = A + B'C$$



A	B	C	V _{out}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Transistor Transistor Logic

It is a logical family consisting completely of transistors. It employs a transistor with multiple emitters. It reduces delay in DTL.

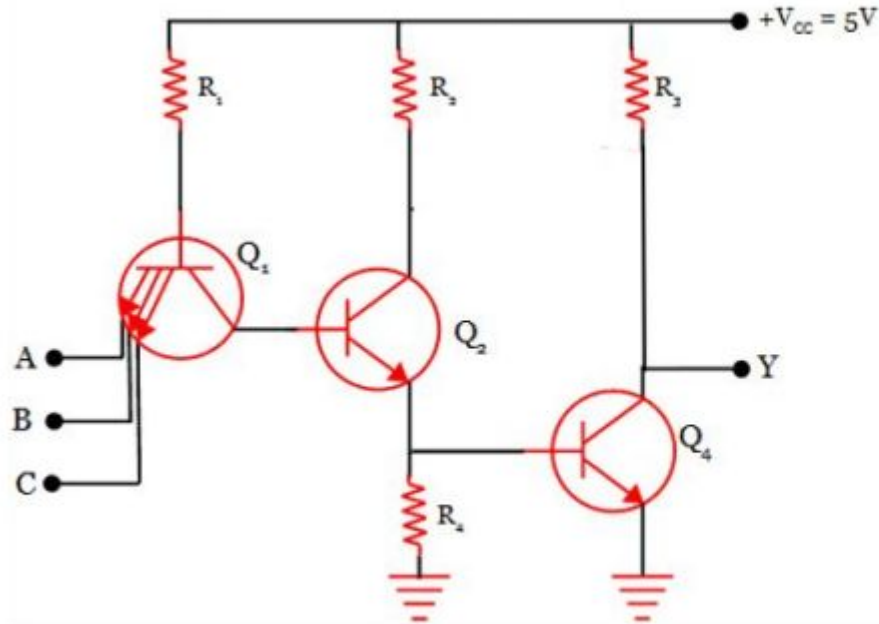
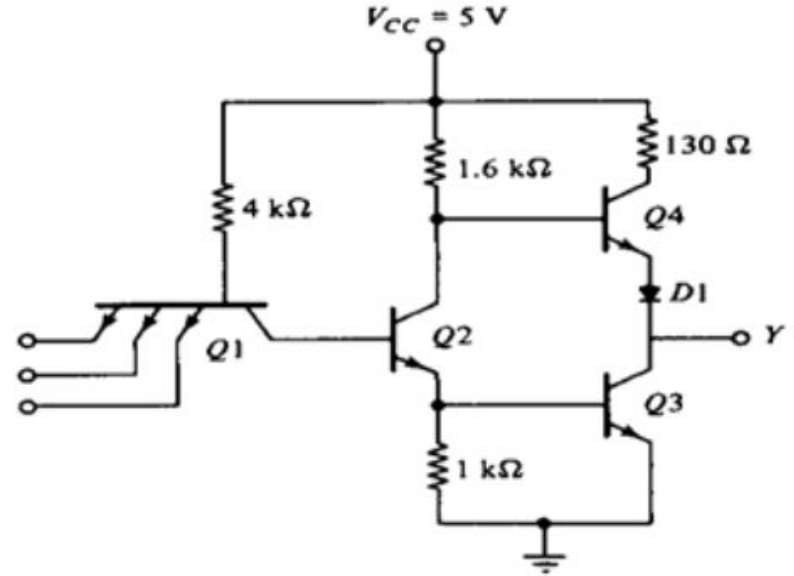


Fig- TTL 3 input NAND gate

A	B	C	V _{out}
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Transistor Transistor Logic

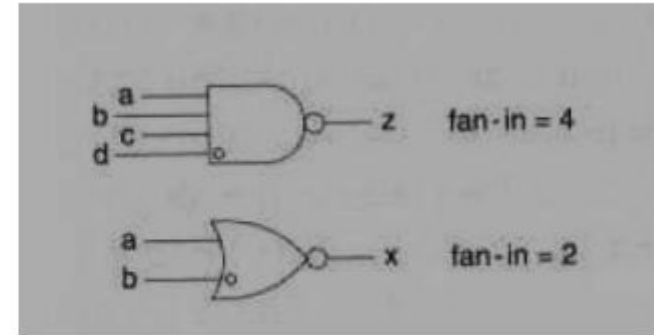
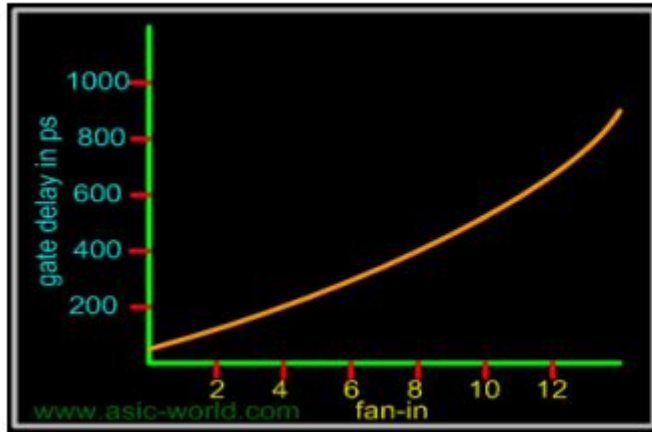
Totem Pole means the addition of an active pull up the circuit in the output of the Gate which results in a reduction of propagation delay.



Totem Pole Output TTL

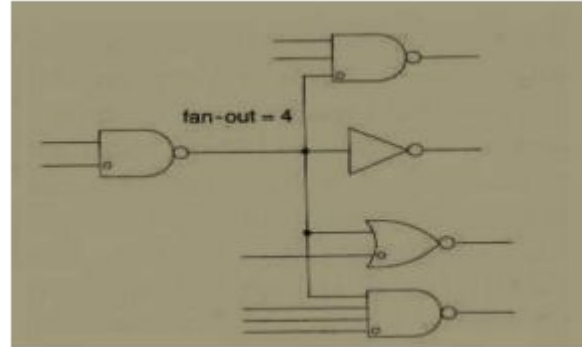
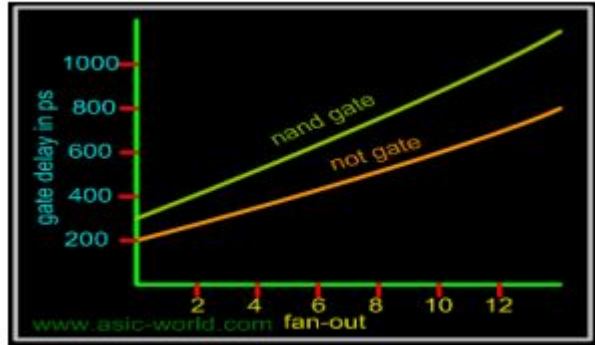
Fan-in and Fan-out

Fan-in is the number of inputs a gate has, like a two input AND gate has fan-in of two, a three input NAND gate as a fan-in of three. So a NOT gate always has a fan-in of one. **It is quadratically related to gate delay.**



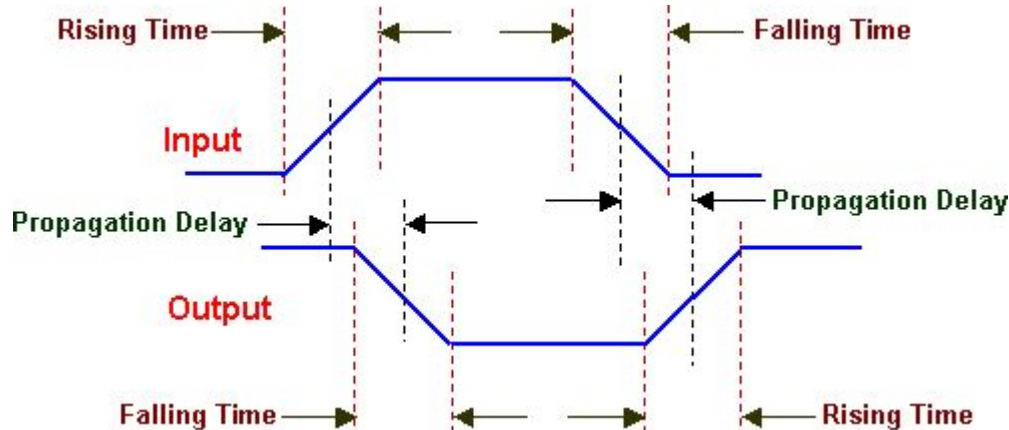
Fan-in and Fan-out

The number of gates that each gate can drive, while providing voltage levels in the guaranteed range, is called the standard **load** or **fan-out**.



Delays in Logic Circuits

- **Rise Time:** Amount of time it takes the output voltage to go from 10% of the Logic "1" level to 90% of the Logic "1" level. Measured in nS.
- **Fall Time:** The time required for the output voltage to go from 90% of the Logic "1" level to 10% of the Logic "1" level. Measured in nS.
- **Propagation Delay:** Transistors within a gate take a finite amount of time to switch. This means that a change on the input of a gate takes a finite amount of time to cause a change on the output.



Power Dissipation

Each gate is connected to a power supply VCC (VDD in the case of CMOS). It draws a certain amount of current during its operation. Since each gate can be in a High, Transition or Low state, there are three different currents drawn from power supply.

ICCH: Current drawn during HIGH state.

ICCT: Current drawn during HIGH to LOW, LOW to HIGH transition.

ICCL: Current drawn during LOW state.

For TTL, ICCT the transition current is negligible, in comparison to ICCH and ICCL.

$$\text{Average Power Dissipation} = V_{cc} * (ICCH + ICCL)/2$$

For CMOS, ICCH and ICCL current is negligible, in comparison to ICCT. So the Average power dissipation is calculated as below.

$$\text{Average Power Dissipation} = V_{cc} * ICCT.$$

Power Dissipation

A certain gate draws 1.8mA when its output is HIGH and 3.2 mA when its output is LOW. What is the average power dissipation if V_{CC} is 5V for TTL.

The average I_{CC} is given as,

$$I_{CC(avg)} = \frac{I_{CCH} + I_{CCL}}{2} = \frac{1.8 + 3.2}{2} = 2.5mA$$

$$P_{D(avg)} = I_{CC(avg)} \times V_{CC} = 2.5mA \times 5V = 12.5mW$$